

Study notes — PTU, *Improved regularity for a nonlocal dead-core problem*

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PROGRESS.md.

The equation studied:

$$-(-\Delta)^s u = u_+^\gamma - u_-^\gamma \quad \text{in } B_1, \quad \gamma \in (0, \tfrac{1}{3}), \quad s \in (\tfrac{1}{2}, 1).$$

Lesson 1 — Orientation

Physical picture (dead cores). A reactant diffuses into a catalyst pellet and is consumed by a reaction of order $\gamma < 1$. If the reaction is fast, the reactant is used up before reaching the center, leaving a region where the concentration is *exactly* zero — the **dead core**. The sublinear, non-Lipschitz term u^γ (with $0 < \gamma < 1$) is precisely what lets u reach zero on a whole set rather than at isolated points.

The equation, unpacked. - $(-\Delta)^s$ = fractional Laplacian: *nonlocal* diffusion. It averages differences $u(x) - u(y)$ over **all** of $\mathbb{R}^n$, weighted by $|x - y|^{-n-2s}$. Far-away values influence x — the root of every technical difficulty. - **Two-phase:** u has no fixed sign. RHS is u_+^γ where $u > 0$, $-u_-^\gamma$ where $u < 0$. The set where u vanishes with derivatives = **branching points** (the free-boundary analogue). - $\gamma \in (0, 1/3)$, $s \in (1/2, 1)$: forced by the method. $\gamma < 1/3 \iff \gamma < \frac{1}{1+2s}$ is the integrability condition that makes a key increment lie in the tail space L_s^1 .

Main result (eq. 1.2). At a branching point x_0 :

$$|u(x)| \leq C |x - x_0|^{\frac{2s}{1-\gamma}}.$$

“Improved” because the exponent $\frac{2s}{1-\gamma} > 2s + \gamma$ beats the Schauder ceiling $C^{2s+\gamma}$.

Strategy (the whole paper in 4 moves). 1. **Scale** around the branching point: $v_r(x) = u(rx)/r^{2s/(1-\gamma)}$, study $r \rightarrow 0$. 2. **Pass to the limit** $v_r \rightarrow v_0$; the reaction term dies in the limit, leaving a clean equation. 3. **Liouville** (§3): the only such limit with the allowed growth is a polynomial of degree $\nu \in \{1, 2\}$. 4. **Contradiction:** polynomial structure forces the sharp growth (1.2).

This skeleton — *scale* $\rightarrow$ *limit* $\rightarrow$ *Liouville* $\rightarrow$ *contradict* — is reused by Correa-dos Prazeres for the p -Laplacian and is what our project ports.

Lesson 2 — The grammar of §2 (spaces, solutions, regularity toolbox)

1. The fractional Laplacian.

$$(-\Delta)^s u(x) = c_{n,s} \text{P.V.} \int_{\mathbb{R}^n} \frac{u(x) - u(y)}{|x - y|^{n+2s}} dy.$$

- **P.V.**: the weight $|x - y|^{-n-2s}$ is non-integrable near $y = x$; symmetric cancellation of the odd part of $u(x) - u(y)$ makes the limit finite. Needs u locally $C^{1,1}/C^2$ at x for the classical value — motivates viscosity solutions. - **Scaling (memorize)**: $(-\Delta)^s[u(\lambda \cdot)](x) = \lambda^{2s}((-\Delta)^s u)(\lambda x)$ — order- $2s$ derivative. This dictates the $r^{2s/(1-\gamma)}$ normalization in §4. - $c_{n,s}$: normalization, depends only on n, s ; ensures $(-\Delta)^s \rightarrow -\Delta$ as $s \rightarrow 1$ (used in §5).

2. Tail space L_s^1 . $\|u\|_{L_s^1} = \int_{\mathbb{R}^n} \frac{|u(y)|}{1 + |y|^{n+2s}} dy$. Finiteness $\iff$ the far-field tail is tame enough for nonlocal diffusion to be defined/stable. **Every difficulty (the §4 gap, the p -port) reduces to whether a rescaled function stays in (the analogue of) L_s^1 .**

3. Viscosity solutions (Def. 2.1). Touching-test-function notion: u is a subsolution if wherever smooth φ touches u from above at x_0 , the operator inequality holds for $v = \varphi$ (in small ball) $\cup u$ (outside). Chosen because (i) never differentiates u directly, (ii) supports comparison/existence, (iii) generalizes to fully nonlinear / p -Laplacian. Note v keeps real u outside the ball — nonlocality built in.

4. Comparison principle (Thm 2.1). $Lu_1 \leq 0 \leq Lu_2$ in B_1 and $u_1 \geq u_2$ outside $B_1 \Rightarrow u_1 \geq u_2$ everywhere. Ordering on the complement (not boundary) controls the inside; powers Perron existence.

5. Hölder seminorms. $[u]_{C^\alpha} = \sup \frac{|u(x) - u(y)|}{|x - y|^\alpha}$; $[u]_{C^{1+\alpha}} = \max_{|\beta|=1} [D^\beta u]_{C^\alpha}$; $[u]_{C^{2+\alpha}}$ likewise with $|\beta| = 2$. §4's θ_k just tracks these under rescaling.

6. Three regularity theorems (black boxes). - **Thm 2.2** (homogeneous): $(-\Delta)^s u = 0$, $\|u\|_{L_s^1} < \infty \Rightarrow u \in C^{1+\sigma}(B_{1/2})$, $\|u\|_{C^{1+\sigma}(B_{1/2})} \leq C(\|u\|_{L^\infty(B_1)} + \|u\|_{L_s^1})$. Tail term is the price of nonlocality. **Engine for the Liouville theorem.** - **Thm 2.3** (bounded RHS): $-(-\Delta)^s u = f \in L^\infty$, $s \in (1/2, 1) \Rightarrow u \in C^{1+\sigma}(B_{1/2})$, any $\sigma < 2s - 1$. Why $s > 1/2$: need $2s - 1 > 0$ to be in the differentiable regime. Baseline regularity. - **Thm 2.4** (Schauder): $(-\Delta)^s u \in C^\sigma \Rightarrow u \in C_{loc}^{\sigma+2s}$. With RHS $\in C^\gamma$ gives $u \in C_{loc}^{2s+\gamma}$ = the **Schauder ceiling** the main theorem beats.

Flow: 2.3 $\rightarrow$ differentiable regime; 2.4 $\rightarrow$ ceiling $C^{2s+\gamma}$; 2.2 $\rightarrow$ engine for Liouville. Main theorem beats the ceiling at branching points $(\frac{2s}{1-\gamma})$.

Lesson 3 — The Liouville theorem (§3), the technical heart

Liouville = “entire solution + controlled growth $\Rightarrow$ polynomial.”

Theorem 3.1. If $s \in (1/2, 1)$, $(-\Delta)^s(u(x+h) - u(x)) = 0$ in B_1 for all h (3.1), and $[u]_{C^{\nu+\alpha}(B_R)} \leq CR^{\beta-\alpha}$ for all $R \geq 1$ with $0 \leq \alpha < \beta < 1$, $\nu \in \{1, 2\}$ (3.2), then u is a degree- ν polynomial.

Why the hypothesis is on increments (3.1), not on u : that's the form the blow-up limit v_0 in §4 satisfies — v_0 isn't s -harmonic, but all its first differences are.

Proof, $\nu = 1$ (five moves): 1. **Subtract the affine part** $w = u - u(0) - Du(0) \cdot x$. Legal because (a) increments of an affine function are constant in x and $(-\Delta)^s(\text{const}) = 0$, so w solves the same increment equation — *pure linearity, the move that dies for $p \neq 2$* ; (b) $[\cdot]_{C^{1+\alpha}}$ ignores affine parts so (3.2) is inherited. Now $w(0) = |Dw(0)| = 0$. 2. **Differentiate + ride growth into the tail:** $w_e = Dw \cdot e$, $w_e(0) = 0 \Rightarrow |w_e(x)| \leq [w]_{C^{1,\alpha}(B_{|x|})}|x|^\alpha \leq C|x|^\beta$ outside B_1 . Since $\beta < 1 < 2s$, $\int_{B_1^c} |w_e|/|y|^{n+2s} < \infty \Rightarrow w_e \in L_s^1$. ($\beta < 1$ is exactly what buys this.) 3. **Stability $\Rightarrow w_e$ is s -harmonic:** $w_e = \lim$ difference quotients of w , each s -harmonic; they converge in L_s^1 globally, so Caffarelli-Silvestre stability [11, Lem 5] passes the equation to the limit: $(-\Delta)^s w_e = 0$ in B_1 . (!) **This lemma needs GLOBAL L_s^1 convergence — here free, but in §4 it's exactly the gap.** 4. **Gradient-decay trick $\rho^{\beta-1} \rightarrow 0$ (transplants to p !):** $v(x) = \rho^{-\beta} w_e(\rho x)$ has $\|v\|_{L^\infty(B_1)} \leq C$ uniformly; Thm 2.2 $\Rightarrow \|Dv\|_{L^\infty(B_{1/2})} \leq C$; but $= \rho^{1-\beta} \|Dw_e\|_{L^\infty(B_{1/2\rho})} \Rightarrow \|Dw_e\|_{L^\infty(B_{1/2\rho})} \leq \rho^{\beta-1} C_3 \rightarrow 0$. So $Dw_e(0) = 0$; translation invariance $\Rightarrow Dw_e \equiv 0$. 5. $w_e \equiv \text{const} = 0$ for every $e \Rightarrow Dw \equiv 0 \Rightarrow w \equiv 0 \Rightarrow u$ affine. $\square$

$\nu = 2$: subtract quadratic Taylor poly, run the same machine on $w_{ee} \Rightarrow u$ quadratic. (This is the case §4 uses, $\beta = \frac{2s}{1-\gamma} - 2$.)

Lemma 3.1 (jet-to-growth, mean value theorem only — transplants verbatim): vanishing jet $u(0) = |D^{\nu-1}u(0)| = |D^\nu u(0)| = 0 + \sup_{0 < r < 1/2} r^{\alpha-\beta} [u]_{C^{\nu+\alpha}(B_r)} \leq A$ ($\beta > \alpha$) $\Rightarrow |u(x)| \leq A|x|^{\nu+\beta}$ in $B_{1/2}$. Converts seminorm control + vanishing jet into pointwise growth; used in §4 to turn the contradiction hypothesis into Liouville's input.

Carry-outs for the project: 1. **Linearity used twice** (affine subtraction; derivative = limit of s -harmonic increments) — both fail for $p \neq 2 \Rightarrow$ must use Correa-dos Prazeres no-subtraction Liouville. 2. **Stability's global- L_s^1 hypothesis** = seed of the §4 gap. 3. $\rho^{\beta-1} \rightarrow 0$ **decay trick survives** to the nonlinear/p setting.

Lesson 4 — §4: the blow-up proof and the gap

Goal: Theorem 4.1, the sharp growth $|u(x)| \leq C|x - x_0|^\alpha$, $\alpha = \frac{2s}{1-\gamma}$, at a **branching point** ($u(x_0) = |Du(x_0)| = 0$). The whole proof is one **argument by contradiction**.

Architecture (memorize this skeleton — the project reuses it verbatim): contradiction $\rightarrow \theta_k$ device picks the right blow-up scale $\rightarrow$ compactness gives a limit $v_0 \rightarrow$ Liouville forces v_0 polynomial $\rightarrow$ vanishing jet kills it $\rightarrow$ contradicts the normalization.

Step 1 — set up. Normalize $x_0 = 0$, $\|u\|_\infty = 1$. Baseline theory: $u \in C^{2s+\gamma}$ (the **Schauder ceiling**, best generic regularity). If the estimate fails, get solutions u_k , branching point 0, uniform $[u_k]_{C^{2s+\gamma}}$, and points x_k with $|u_k(x_k)| > k|x_k|^\alpha$ (4.3) — the violation grows with k .

Step 2 — the θ_k device (the engine).

$$\theta_k(r') := \sup_{r' < r < 1/2} r^{2s+\gamma-\alpha} [u_k]_{C^{2s+\gamma}(B_r)} \quad (4.4).$$

The prefactor exponent $2s + \gamma - \alpha < 0$ (since $\alpha > 2s + \gamma$, “better than Schauder”). The violation (4.3) + jet-to-growth (Lemma 3.1) force $\lim_{r' \rightarrow 0} \theta_k(r') > k$. *Why:* if θ_k stayed bounded, the seminorm would decay at the rate that **implies** the good growth $|u_k| \leq C|x|^\alpha$, contradicting x_k .

Step 3 — blow-up scale r_k . Pick $r_k > \frac{1}{k}$ nearly achieving the sup: $r_k^{2s+\gamma-\alpha} [u_k]_{C^{2s+\gamma}(B_{r_k})} \geq \frac{1}{2} \theta_k(r_k)$ and $\geq \frac{1}{k} \theta_k(1/k) \rightarrow \infty$. Forces $r_k \rightarrow 0$ (zoom into the branching point).

Step 4 — the blow-up v_k . $v_k(x) := \frac{u_k(r_k x)}{\theta_k(r_k) r_k^\alpha}$ (4.6). Two normalizations chosen to hold together: - **Seminorm bound (4.8):** $[v_k]_{C^{2s+\gamma}(B_R)} \leq R^{\alpha-2s-\gamma}$, **only for** $1 \leq R \leq \frac{1}{2r_k}$ — flag this, it is the source of the gap. - **Surviving normalization (4.10):** $[v_0]_{C^{2s+\gamma}(B_1)} \geq \frac{1}{2}$ (limit is nontrivial).

The **anchor trick** ($\int \eta D_{ee} v_k = \int D_{ee} \eta v_k \leq C_n$ pins a 2nd derivative at one point) + the seminorm bound give $|D_{ee} v_k| \leq C R^{\alpha-2}$, hence **compactness:** $v_k \rightarrow v_0$ in $C_{loc}^{2s+\gamma}$.

Step 5 — increments + the $\gamma < 1/3$ condition. Pass to increments $w^h = v_k(\cdot + h) - v_k$ (first differences, lower the order by one). Mean value theorem + (4.8): $|w^h(x)| \leq C_h |x|^{\alpha-1}$ for $|x| \geq 1$ (4.11). This is in the tail space L_s^1 **iff** $\gamma < \frac{1}{1+2s}$ — implied by $\gamma < 1/3$. **This is the entire reason for $\gamma < 1/3$:** the integrability of the increment’s tail. (= your project’s tail condition C1.)

Step 6 — limit passage (4.12) and THE GAP. Since u_k solves the equation,

$$-(-\Delta)^s(v_k(\cdot + h) - v_k) = (\text{prefactor} \rightarrow 0) \cdot [\text{RHS difference}] \quad (4.12),$$

and PTU pass to the limit via the **stability lemma** (needs *global* L_s^1 convergence) to get $(-\Delta)^s(v_0(\cdot + h) - v_0) = 0$ in $\mathbb{R}^n$. **The gap:** (4.11) holds only for $|x| \leq \frac{1}{2r_k}$ (it rests on 4.8); beyond that v_k is merely bounded by $2(\theta_k r_k^\alpha)^{-1}$, and the far-field tail contributes $\sim \theta_k^{-1} r_k^{-\frac{2s\gamma}{1-\gamma}}$, bounded only under a **lower bound on θ_k** that (4.5) does not supply. CS Remark 4.4 shows the global hypothesis cannot be weakened $\Rightarrow$ genuine question for R.T. (your (c’) repair fixes it by **truncating** at $\rho_k = \frac{1}{4r_k}$).

Step 7 — Liouville finishes. v_0 satisfies the increment-form hypothesis (3.1) with $\nu = 2$ (vanishing 2-jet at the branching point), growth $\beta = \alpha - 2 < 1$. Theorem 3.1 $\Rightarrow v_0$ is a degree- ≤ 2 polynomial; vanishing 2-jet $\Rightarrow v_0 \equiv 0$, contradicting (4.10). $\square$

Carry-outs: 1. Skeleton is reused in the project: swap $C^{2s+\gamma} \rightarrow C^{1,\alpha_0}$, linearity $\rightarrow$ homogeneity. 2. $\gamma < 1/3 =$ the L_s^1 integrability of the increment = tail condition **C1**. 3. The gap is at (4.12); the far-field needs a θ_k lower bound the device doesn't give — **(c') truncation repair**.

Glossary — every recurring term, in plain language

Read this whenever a term in the notes or in `main.tex` is unfamiliar. Each entry gives the intuition first, then the precise meaning, then where it shows up.

Dead core. The interior region where the solution u is *exactly* zero. Physically: where the reactant has been completely consumed. It exists because the absorption u^γ with $\gamma < 1$ is non-Lipschitz at 0.

Branching point / free-boundary point. A point x_0 on the edge of the dead core where u touches zero *with* a horizontal tangent: $u(x_0) = 0$ and $Du(x_0) = 0$. These are the points at which the sharp growth estimate is proved. (“Free boundary” = the boundary $\partial\{u > 0\}$, unknown in advance; “branching” because the two phases $\{u > 0\}$ and $\{u < 0\}$ can meet there.)

Jet (1-jet, 2-jet). The Taylor data of u at a point. The k -**jet** is $(u(x_0), Du(x_0), \dots, D^k u(x_0))$. The **1-jet** = $(u(x_0), Du(x_0))$ is value + gradient (the best affine approximation). A **vanishing 1-jet** means $u(x_0) = |Du(x_0)| = 0$ — exactly the branching-point condition. A **vanishing 2-jet** adds $|D^2 u(x_0)| = 0$ (Hessian zero too); this is the $\nu = 2$ case, available only at $p = 2$.

ν (**nu**). The order of vanishing assumed at the branching point: $\nu = 1$ uses a vanishing 1-jet, $\nu = 2$ a vanishing 2-jet. For $p \neq 2$ only $\nu = 1$ is usable (quadratics aren't (s, p) -harmonic).

Fractional Laplacian $(-\Delta)^s$. Nonlocal diffusion of order $2s$. $(-\Delta)^s u(x) = c_{n,s} \text{P.V.} \int \frac{u(x) - u(y)}{|x - y|^{n+2s}} dy$ — averages how $u(x)$ differs from u everywhere else.

Fractional p -Laplacian $(-\Delta_p)^s$. The nonlinear ($p \neq 2$) version: replace the difference $u(x) - u(y)$ by $J_p(u(x) - u(y)) = |u(x) - u(y)|^{p-2}(u(x) - u(y))$. Reduces to $(-\Delta)^s$ when $p = 2$.

s -harmonic / (s, p) -harmonic. A function with $(-\Delta)^s u = 0$ (resp. $(-\Delta_p)^s u = 0$) — the nonlocal analogue of “harmonic” ($\Delta u = 0$). These are the “trivial” solutions the Liouville theorem classifies.

P.V. (principal value). A symmetric way to make sense of the singular integral defining the operator (the weight $|x - y|^{-n-2s}$ blows up at $y = x$); the odd part cancels, leaving a finite value.

Tail / L_s^1 / Tail $_{p-1,sp}$. Bookkeeping for the far-field. Because the operator is nonlocal, it only makes sense if u doesn't grow too fast at infinity. $\|u\|_{L_s^1} = \int \frac{|u(y)|}{1+|y|^{n+2s}} dy$ (the $p = 2$ version); Tail $_{p-1,sp}$ is its p -analogue. “Finite tail” = “far-field tame enough for the theory to work.” Almost every difficulty in the project is a tail question.

Seminorm $[u]_{C^\alpha}$, **Hölder space** $C^{k,\alpha}$. Measures of smoothness. $[u]_{C^\alpha} = \sup_{x \neq y} \frac{|u(x) - u(y)|}{|x - y|^\alpha}$ quantifies α -Hölder continuity; $C^{k,\alpha} = C^{k+\alpha}$ means k derivatives, the k -th being α -Hölder. $[u]_{C^{1+\alpha}}$ controls the gradient's Hölder norm.

Viscosity solution. A notion of solution needing no derivatives of u : test against smooth functions touching u from above/below. Used because solutions may not be twice differentiable, and because it extends to nonlinear/ p operators.

Weak solution. The other notion of solution, defined by integrating against test functions ($u \in W^{s,p}$). For this problem weak $\equiv$ viscosity (Korvenpää-Kuusi-Lindgren).

Blow-up / rescaling. Zoom in at a branching point: $v_r(x) = u(rx)/r^\alpha$, study $r \rightarrow 0$. The exponent α is chosen so the rescaled functions stay balanced; their limit v_0 is a clean model solution.

Homogeneity (of the operator). $(-\Delta_p)^s(\lambda u) = \lambda^{p-1}(-\Delta_p)^s u$ for $\lambda > 0$ ($(p-1)$ -homogeneity), and the order- sp spatial scaling. In our project these two together make the absorption term vanish along the blow-up — *this replaces the differentiation step that needs linearity*.

Liouville theorem. “Entire solution + controlled growth $\Rightarrow$ trivial (a polynomial / zero).” The engine that classifies the blow-up limit v_0 and forces the contradiction. The project’s “load-bearing wall.”

Increment / first difference w^h . $w^h(x) = u(x+h) - u(x)$. Differencing lowers the growth order by one and, for $p \neq 2$, turns the nonlinear equation into a *linear* one for w^h (with a u -dependent kernel). **Second difference** $\Delta_h^2 u = u(\cdot+2h) - 2u(\cdot+h) + u(\cdot)$ lowers it by two.

Cocycle identity. The bookkeeping rule $w^{h_1+h_2} = w^{h_1}(\cdot+h_2) + w^{h_2}$ for increments; used to show a quantity depending on h is linear in h .

Stability lemma. The theorem letting you pass an equation to the limit: if $u_k \rightarrow u$ (with global tail control) and $(-\Delta)^s u_k = f_k$, then $(-\Delta)^s u = \lim f_k$. The hypothesis that the convergence be *global* (in L_s^1) is the subtle one — the source of the §4 gap.

θ_k **device.** The contradiction machine (PTU §4 / CP §6): assume the estimate fails, build a sequence whose rescaled blow-up survives a normalization but must vanish by Liouville. $\theta_k(r')$ measures the worst seminorm growth across scales.

Gradient anchor. A trick ($\int \eta D_e v = \int D_e \eta v \leq C$) that pins the gradient at one point, giving local bounds on v_k *without* normalizing it in L^∞ — useful nonlocally.

Schauder estimate / “better than Schauder.” Schauder theory gives the generic a-priori regularity ($C^{2s+\gamma}$ for $p = 2$). “Better than Schauder” = the growth at branching points is *smoother* than that generic ceiling — the whole point of the word “improved” in the title.

Nondegeneracy. Two uses. (1) Lower bound matching the growth: $\sup_{B_r} |u| \geq c r^\alpha$ (makes the estimate sharp). (2) For the $p \neq 2$ Liouville: that the limit v_0 has $Dv_0 \not\equiv 0$ on open sets, so the increment’s v_0 -dependent kernel stays uniformly elliptic.

Admissible region. The set of parameters (s, p, γ) for which the theorem's hypotheses hold simultaneously — the analogue of PTU's $\gamma \in (0, 1/3)$, $s \in (1/2, 1)$. Determined by intersecting the constraint conditions (C1 tail, C2 the cap $\alpha < 2$, etc.).